

Základy bioinformatiky – zápočtový domácí úkol

Vypracovala: Karolína Šustrová	Kruh: 330	Skupina: čtvrtek 8:00
Zadaná rodina glykosidhydroláz:	GH92	

Jedná se o enzymy z rodiny α -mannosidáz vyskytující se v bakteriích *Microbacterium* sp., *Bacteroides thetaiotaomicron*, *Cellulosimicrobium cellulans* a dalších. Tyto enzymy jsou Ca^{2+} -dependentní α -mannosidázy, které katalyzují hydrolýzu glykosidových vazeb prostřednictvím mechanismu jednoduchého přemístění vedoucího k inverzi anomerní konfigurace a produkují tedy β -mannózu. Vykazují dvoudoménnou strukturu složenou z malé N-terminální domény a velké C-terminální domény. V komplexu s inhibitory je Ca^{2+} ion koordinován na 2- a 3- OH skupinu inhibitoru.

Pro zarovnání a vytvoření fylogenetického stromu jsem vybrala 17 sekvencí, které odpovídaly enzymům α -1,2-mannosidáza, α -1,3-mannosidáza, α -1,4-mannosidáza, mannosidáza Man1, mannosidáza Man2, mannosidáza Man4, mannosidáza Man5 a exo- α -1,2-mannosidáza čtyř různých mikroorganismů.

Sekvence jsem si nejdříve našla na UniProt a stáhla si je přes ENA ve fasta formátu. Pro mnohočetné zarovnání jsem použila nástroj Clustal Omega a vizualizaci zarovnání jsem vytvořila pomocí ESPript 3.0 nástroje, kam jsem vložila soubor se zarovnáním z Clustalu. Fylogenetický strom jsem vytvořila pomocí nástroje Simple Phylogeny, kam jsem zadala zarovnané sekvence v clustal formátu.

Ze zarovnání jsem zjistila, že sekvence jsou si podobné nebo dokonce shodné ve velkém množství krátkých úseků především v jejich střední části a že se navzájem liší svojí délkou.

Zarovnání sekvencí:

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AFK83609.1 .....
AAO78889.1 .....
AAO77736.1 .....
AAO79067.1 .....
AAO79095.1 .....
BAA76709.1 MTVPFRPRNERSAR.....RLWA.AAGAALTVTVMAMLT.....PVTASAAAAGGT
AFK83608.1 MTRPLPPGRAVARSGSGRARPLGLVLA..AALAVPLGVPLAAPAGALAA.APAAAAEPGD
CAD7619346.1 .....MKKRAIQKIISIVSLAVLTSVPVMQKPSVLAASSDSTSASKTD
AFK83605.1 .....
AFK83606.1 .....
AAO78636.1 .....
AAO79198.1 .....
AAO79178.1 .....
AAO79070.1 .....
AAO78236.1 .....
AAO76985.1 .....
AAO78878.1 .....
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AFK83609.1 .....
AAO78889.1 .....
AAO77736.1 .....
AAO79067.1 .....
AAO79095.1 .....
BAA76709.1 FASSFESTDATPVLTG....TGQPVNVTGSR.....FVPGSMLPAIAPDGVT
AFK83608.1 FSSSFESGDPAALPTTVAERDGAQWQANVGSFTAGL.....PGSVLGQ..LKGVT
CAD7619346.1 FFSSFEKSDLQLTWTINTVETDA....NGKKMSSGIDGNVVRDLILGDIIDK..VVQVTA
AFK83605.1 .....
AFK83606.1 .....
AAO78636.1 .....
AAO79198.1 .....
AAO79178.1 .....
AAO79070.1 .....
AAO78236.1 .....
AAO76985.1 .....
AAO78878.1 .....
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AFK83609.1 .....
AAO78889.1 .....
AAO77736.1 .....
AAO79067.1 .....
AAO79095.1 .....
BAA76709.1 SAQNTFNEAAAFADANASTKKWLAFARTGWIQYQLTAAQPMKYTLTSGGDAEARDPKNF
AFK83608.1 SAQNLFPNEGAANLADGSSGTTKWLAFASTGWVRYEFAEPVSFVAYTMTSGDDAAGRDPK
CAD7619346.1 SANNFPNEIDSKLIDGDPPTTKWLAFEFPTANIVLKLAEFVAVVKYALTSANDAKGRDPKNW
AFK83605.1 .....
AFK83606.1 .....
AAO78636.1 .....
AAO79198.1 .....
AAO79178.1 .....
AAO79070.1 .....
AAO78236.1 .....
AAO76985.1 .....
AAO78878.1 .....
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AFK83609.1 .....
AAO78889.1 .....
AAO77736.1 .....
AAO79067.1 .....
AAO79095.1 .....
BAA76709.1 RVGGSNNGTDWTTVDERSGEMFTGRQQTRTFELAAPSDPYLYYRLDVLETRTANTGIVQL
AFK83608.1 TVEGSNDGSTWAALDRRTDEDFPNRQQTRTFELEAPTAAYTYLRLNVTAN..SGDSIVQL
CAD7619346.1 TLYGSLDGTNWTAVDTREGEDFKDRFQRNMYDLKN.TTKYLYYKLDITKN..AGDSITQL
AFK83605.1 .....
AFK83606.1 .....
AAO78636.1 .....
AAO79198.1 .....
AAO79178.1 .....
AAO79070.1 .....
AAO78236.1 .....
AAO76985.1 .....
AAO78878.1 .....
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AFK83609.1 .....
AAO78889.1 .....
AAO77736.1 .....
AAO79067.1 .....
AAO79095.1 .....
BAA76709.1 ADFEP I A.VSDGVFVADDLRLAIGNGFTASDTAKTGVGFTGTAKALQYSGRHLAAGPASST
AFK83608.1 AGWDL S A D L S A . G P S A A P M T T K V G T G P R V S F T N K A G V G F S G L H S L R Y D G S H L A D G E T Y A T
CAD7619346.1 AEISLS D G I E V P A P P P G D M K S L I G K G P T S S Y T A K T N V G W T G L G A L N Y S G T H L S D G R A Y S Y
AFK83605.1 .....
AFK83606.1 .....
AAO78636.1 .....
AAO79198.1 .....
AAO79178.1 .....
AAO79070.1 .....
AAO78236.1 .....
AAO76985.1 .....
AAO78878.1 .....

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AFK83609.1 .....
AAO78889.1 .....
AAO77736.1 .....
AAO79067.1 .....
AAO79095.1 .....
BAA76709.1 TTVYSDVDVAVEKGESELSYLVFPVLDGE..QTYAATFVAVD LTFDDGTTLSTSGAVDSYG
AFK83608.1 NVLYDDVDVVVGEDTRLSTIFFPELLDD..LQYPSTYAAVDV LFTDGTLYLSDLGARDAHE
CAD7619346.1 NKLYD.VDILVTPATELSYFIAPEFTDKNHN DYSSYVSVDLAFSDGTYLHDLKAVDQYG
AFK83605.1 .....
AFK83606.1 .....
AAO78636.1 .....
AAO79198.1 .....
AAO79178.1 .....
AAO79070.1 .....
AAO78236.1 .....
AAO76985.1 .....
AAO78878.1 .....

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AFK83609.1 .....ALA
AAO78889.1 .....MK.....THFS
AAO77736.1 .....MK.....THFS
AAO79067.1 .....MN
AAO79095.1 .....
BAA76709.1 YAANARKQGQSNALWPNQWNKVTVDLGQF.EGRTID DILFTYDHPGADVKGIEVPTAATS
AFK83608.1 TVATAQAQGEKILYADQWNSVRVDLGDVAEGKTVDQVLLGYDNPGGH.....AGTK
CAD7619346.1 VGLNPKDQGD SKYLYVNQWNTIKSTIGSVAAGKTIKRILVAYDNPKGP.....G..A
AFK83605.1 .....MRR..PR.....LALL
AFK83606.1 .....MSL
AAO78636.1 .....
AAO79198.1 .....
AAO79178.1 .....MY
AAO79070.1 .....
AAO78236.1 .....
AAO76985.1 .....MIIMKRKSPV.....MAVA
AAO78878.1 .....MN

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AFK83609.1 VVGLA.....P.....ATAASAAPEPPSADYASLVDFVGTET.....GDF
AAO78889.1 FKHL L F I G .....GAVLYSMQISAVKNPVDYVSTLVGTQSKFE.....LST
AAO77736.1 FKHL L F L G .....GAVLYSLQSSAVKNPVDYVSTLVGTQSKFE.....LST
AAO79067.1 KQKLLSITM.....TLLGVSSIFAQKQPVDDYVNPLMGTDSKIS.....LSN
AAO79095.1 MKRLLL..I.....ACILNCTLLSQAKDWTQYVNPLMGSSQSTFE.....LST
BAA76709.1 FSGW..LDD.....VQIGTATARDISDGLVSYVDTRRGTNSTGG.....FSR
AFK83608.1 FAGW..LDD.....VEITAEPATIDGSSLANYYDTRRGTLASGS.....FSR
CAD7619346.1 FRGS..IDD.....IKIDGKFPVQKAFGSPIDYVNILRGTQSNGS.....FSR
AFK83605.1 AAGLALAVAPGTL.LPVAAGAAPADEGTVTAAAGDDLTLEVNPFVGTET.....SE
AFK83606.1 ALPLAAYAAPGIGASPATAAGTEAATGSDAAAVDGPLVDYVNPFVGTET.....DD
AAO78636.1 .....MTPSVAQNTKYVNLFVGTET.....GDN
AAO79198.1 .MKHKFLFIL.....LFSLVLEGMV.TTQAVAGDYVHQVNTLVGTET.....GTGLTS
AAO79178.1 KASANILIAC.....SFLWLTA C S S A G V T E D L I P T D Y V N P F I G A S T S V G A A G V Y H G L
AAO79070.1 .....MAQTEKLTDYVNPFVGTET.....GY
AAO78236.1 .MKKLLLSVC.....AFSLTLA.....TLQAGEITKYVNPFVGTETGAIDG.....GLS
AAO76985.1 LAGLFMYTSC.....SPTEQAETKDYTYVNNTFVGTETGAA.....DN
AAO78878.1 AFKLLMAASC.....FLGLATSGVAKQDYKSEGLLQYVDPYVGTETGSG.....FH

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40      50      60      70      80
AFK83609.1 GNDMPAAQAPNGLAKVNPRTTP...GRNNTGYDYAQ.....SKISGFTHINLDGVGGS
AAO78889.1 GNTYFATALPWGMNFWTPQTGK.M..GDGWAYTYNA.....DKIRGFKQTHQPSPWM.
AAO77736.1 GNTYFATALPWGMNFWTPQTGK.M..GDGWAYTYDA.....DKIRGFKQTHQPSPWM.
AAO79067.1 GNTYFAIALPWGMNFWTPQTGK.M..GDGWAYTYAS.....DKIRGFKQTHQPSPWI.
AAO79095.1 GNTYFAIARPWGMNFWTPQTGK.M..GDGWQYTYTA.....NKIRGFKQTHQPSPWI.
BAA76709.1 GNNLFAATAWPNGFNFITPMTNA.D..NTGTIYQYQRANNAQNKIGLNGIGFSHQPSIWM.
AFK83608.1 GNNIFATATPNGFNFWTPYTNA.S..SQSWLYEYHKANNANNKPVLOQFGISHEPSPWM.
CAD7619346.1 GNNFPAVAIPHGFNFWTPYTNA.G...SSWIYQYHESNSVNNLPQIQAFSVSHEPSPWM.
AFK83605.1 GNAFYFGATVPFGMVQLSPDNTN...SYASTSYSTNA.....GRVWGFSHRHVNSAGC.
AFK83606.1 GNTYFGAAVFPFGMVQLSPDNH...NV...GYDYDR.....TSVRGFSLVHLSGVGC.
AAO78636.1 GQVAFGAAPFGMVVCVCPDND....PRSHAGYDYAV.....TKVSGISVNRLSGVGCS
AAO79198.1 GYLTFGATYPYGMVQFTPSYF.....SKRSGFVINQLSGGGC.
AAO79178.1 KKTFFGATTPYGMVQVSPNTI...TGDNSSGYSDHEH.....KTIEGFAFTQMSGVWGF
AAO79070.1 GNVYFGAQIPFGGIQISPDTD.SRFYDAASGYKYNH.....LTLMGFSLTHLSGTGI.
AAO78236.1 GNNYFGATSPFGMIQLSPDISEAPNWGDASGYDYNR.....NTIFGFSTRLSGTGA.
AAO76985.1 GHTFFGACYFFGMIQTSFVTG.AVGWRYCSEYTYQD.....SLIWGFQTHLNGTGC.
AAO78878.1 GHVFGVGTSVFYGMVQLGPNNI.HKGWDWCSGYHYSD.....SILTFGSHTHLSGTGC.
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90      100     110     120     130     140
AFK83609.1 GGGDLLVVPISGSYTARPG....TGTYAHPFSHDDDAAGFGFYSVGLGNVAGTDGAIT
AAO78889.1 NDYQQFSIMPIITG...GL...VFDQDQRASWFSSHKAQVAKPYKYKVYLAD.....
AAO77736.1 NDYQQFAIMPIITG...GL...VFDQDQRASWFSSHKAQVAKPYKYKVYLAD.....
AAO79067.1 NDYQQFSIMPMTK...QL...KIDQDSRASWFSSHKAQVAKPYKYKVYLSE.....
AAO79095.1 NDYQQFSIMPIVIG...QP...VFDEEKRASWFAHKGEVATPYKYKVYLAEL.....
BAA76709.1 GDRNQLAVLPAAT..SNP...TSNLNDRKLTFSHENEIARPDIDVITF.D.....
AFK83608.1 GDRNQLTFLPSTA.SGTP...DALSTRGLEFDHADETARPDYGVTF.T.....
CAD7619346.1 GDRQTFQVMPASASTAATP...NANRDSRALEFNHANEIAQPHYYSVKF.E.....
AFK83605.1 PAAGELLVTPDTSATP....RTSRSFIAIKDKQKSTERASAGFYEVTLA.....
AFK83606.1 GLGGFLPTLPTTGAIT....STDYGYALGFSSHDDDAAGFGFYRVVGLQAP.....
AAO78636.1 GGGGNLIRPVAPSQ.....ELHIKKSRKATPGYYSTAF.T.....
AAO79198.1 EHMGNFPTFPVKGKLMSPDNI....LNY..RINVSEKKGHAGYEAAMV.Q.....
AAO79178.1 GDLGNFLVMPITGELQKIAGKEDGSIKGYRSSYDKATETAKAGYYSVELTD.....
AAO79070.1 PDLGDFLFIPGTGEMKLEPGTHEDPDQGYRSRYSHDKQWASPNYYAEVLAD.....
AAO78236.1 SLDLIDITLMPITSSG.RT.....SSAFTHDEKARFGYYSQVMLKD.....
AAO76985.1 MDLGDILVMPVTGT.R.....ARAWDAYRSHFPKDKAATPGYYSVELSD.....
AAO78878.1 TDLGDILIMPLNNEI.RTPRGNQDDIRDGYASKYSHDNIEARPEYYSVELLDR.....
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150     160     170     180     190
AFK83609.1 GAPGTIEAEVAAATRSQGVHRYAFFAGSTPSLVVDLEINNTSR.....RSSSVQV
AAO78889.1 ...HDVTTTELVPITERAAMFRFTYPEETKNAYVVIDAFDKGS.....YVKV
AAO77736.1 ...HDVTTTELAPTERAVMFRFTYPEETKNAYVVIDAFDKGS.....YVKV
AAO79067.1 ...YNMTTEIAPTERCAYFRFTYPEASDAYVVVDAFDRGS.....YVKV
AAO79095.1 ...HDIVTEMTPTERAVLFRFTYPEENDHSYVVVDAFDKGS.....YIKI
BAA76709.1 ...NGIRTEVTAIDHGAIRFEFAGDAS.SLIDQLVNSS.....KLTII
AFK83608.1 ...NGSAIEATPTDHGAVLRFSYPGAKG.HVLVDKVDGSS.....KLTII
CAD7619346.1 ...NGIRTEMTPTDHAAMFKFTFTGATS.NLIFDNVNNNG.....GLTII
AFK83605.1 ...NDVHAELTATIRVGAHRYTTFPASTTSHLSFNVGQTLR.....DAGASSVTW
AFK83606.1 ...AGTIEAELTATERTGVQRYTFFATAQANVLLNAGQALN.....RVTESDVRV
AAO78636.1 ...NGIKTELATINAMAVERYKFFRSLSAALWIDFASTFE.....DVATCHYKR
AAO79198.1 ...EDIKAKLTVTERTGMASYEYPAADQQYGTIIIGG.GIS...AT...PIEQAAIVI
AAO79178.1 ...YKIKVESSATPHCGILQFTTFPNEQSRQIDILARRVGGTST.....SQYVKV
AAO79070.1 ...YGVKAEMTSGVRSGMFRFTYPEESDNAFIMIDMNHTLW...QSCE...WSNLRM
AAO78236.1 ...ENINAELTTIQRNGIHRYYQYPAGKDAEILIDMDHSADKGSWGRIINSQIR...I
AAO76985.1 ...PQVKAELTASIHAAALHRYTYHKADSSALLIDLQHGPA...WREQYHSQVNSCEVNW
AAO78878.1 ...YNIKAEELTATDRVGFHRYTYPEGKEFVSLLIDLREGNG...SNA...YDSYIRKV...
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200     210     220     230
AFK83609.1 ...ETRADGTVEL...SGQVTGYFYNAAYLYYTARTLQPAT.VQTWGDDDRDLV....
AAO78889.1 IPEENKIIIGYSTK...NSGGVPE....NFKNYFVIOFDKPFFTFTSGVKENNILP....
AAO77736.1 IPEENKIIIGYSTK...NSGGVPE....NFKNYFVIOFDKPFFTFTSVFENNILP....
AAO79067.1 IPEENKIIIGYSTK...NSGGVPE....NFKNYFVIOFDKPFFTFNKVWADYHLE....
AAO79095.1 IPEENKIIIGYTTR...NSGGVPE....NFKNYFVIOFDKPFITYKAIVENGNLQE....
BAA76709.1 SG...DIVSGWVDG...SGGW.PG....RTRMFVYGTDRAPATAGATTV.....
AFK83608.1 DQATGTISGWVEN...SGLSVG....RTRMFVAGTDRSPTAVGTAAG.....
CAD7619346.1 DAKSGEITIGSDV...KSGLSTG....ATRLFVYAAFDKPKVIKSGKLTG.....
AFK83605.1 ...VDDRTLLEGWVDN...GFCGG...TPDKQRYFFSATDRPVASSSGWGTGDARYVAG...
AFK83606.1 ...VDDRIVETRITV...RGFCQD...TEP.QTIWTRITDRPFVAHGWTGDQVVTAGA...
AAO78636.1 ...ISETCIEGYVQA...KNVCG....HGCKLYFSLNTSQPFQ.....
AAO79198.1 ...TAPNKCEGYAEG...GNFC...GLRTPYKYVFAEFDTDAFETGTWKREELMP....
AAO79178.1 ...LDYTIQGWMMKCTPDGGGWGNG.EGNSDYTVYYYAQFSKPLSNYGFWSADIPDEWVRK
AAO79070.1 ...INDSTITGYKL...KGWP....ERHVYFTATFSKKLTGLRFVQDKKPYIYNTS.
AAO78236.1 ...LNDHAVEGYRII...TGWAK....LRKIYFYMEFSSPILTSTLRDGGRVHE....
AAO76985.1 ...EDAQTLAGHVNN...TVWVD...QDYFFVMKFNRPVIDSL.Y...LPM....
AAO78878.1 ...DDYTVVEGYRYV...RGWSP....SRKVFVVKSDQKIEKFTAYDNTPKL....
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	240	250	260	270
AFK83609.1	DATAQDGVDTGAI	LT	FD	..A.DAGEIGLQVTLSPVSV
AAO78889.1	NETEVQGNHTGAI	IG	FAT	..Q.KGEIVHARVASFSISY
AAO77736.1	NETEAKGNHTGAV	IG	FAT	..K.KGEIVHARVASFSISY
AAO79067.1	THLELQSNHVGA	IG	FST	..K.KGEQVHAKVASFSISY
AAO79095.1	NVAEQITDHAGAI	IG	FKT	..R.KGEQVNAIASSFSISY
BAA76709.1	GDRNGTARYA	AF	DT	..T.TDRTVELRLSSFSISQ
AFK83608.1	NRAD..ARF	AT	FET	..S.SDKTVELRVATSFISL
CAD7619346.1	ESRNNVTGY	VR	FD	ISKD.EDKVVTMKIATSLIS
AFK83605.1	STTSEVAGGNGAV	AV	FD	..T.TDRDVEVSVGVSFVS
AFK83606.1	DAASGGEGRRGAY	VT	FD	..TGGDLDVEAVTAMSYVGAD
AAO78636.1	LEEQKETTAC	LT	FGK	..K.VRSVEVRIGLSALS
AAO79198.1	NTTFAEGEYSGVY	FT	FDV	..N.KKKNIQYKIGVSVSV
AAO79178.1	DEVVSIPYLTRISQAPVIKDKKELEGKHLGFF	TE	FT	..K.EGEQVEMKVGISFVDM
AAO79070.1	RFRSSYEAWGNLMAC	IS	FD	..K.AGEEVTVKTAISVST
AAO78236.1	NTAVINGTDLHGC	FR	FG	..L.NGKPLTKVVALSSVME
AAO76985.1	GETEKGRKIAT	FD	L	..K.PGDELMMKVVALSTTS
AAO78878.1	WDQLKVESVKS	LT	F	..G.NVKEVKIKVAISSVSC

	280	290	300	310	320	330
AFK83609.1	RIDQVVELG.D...	LSFDA	IRDRTRAEW	NATLGRV	ATDASTATDPTGELQRL	FYTHLY
AAO78889.1	ELNLK.ELG.K...	DSFDQ	LVTGKXDIW	NREMSKV	DVEDDN....IDNLR	FYSCLY
AAO77736.1	ELNLK.ELG.K...	NSFDQ	LVANGREI	WNREMSKI	EIEDDN....IDNLR	FYSCLY
AAO79067.1	ELNLK.ELG.N...	KTFEQ	TKAEGRKAN	DVLRGI	KVEDDD....ENRMR	FYSCLY
AAO79095.1	AANMN.ELG.K...	DNIEQ	LAKQKDAW	NQVLGKI	EVEGGN....LDQYRT	FYSCLY
BAA76709.1	KKNHAFVEG.G...	VSFDA	AHAADVDAW	NERLSV	HDVKGAS....DQLVTL	YSSTLY
AFK83608.1	RKNLDLEVT.G...	KTFTE	VKAAQA	WNRDLGVI	EVEGAS....EDQLVT	LSNLY
CAD7619346.1	KKNLEQELGLN...	DTFEG	LKEKAKTE	WNKKLGI	EVEGAS....EDQLVT	LSNLY
AFK83605.1	RANREAEAT.DEGGQVAF	DTV	REEARDAW	NAELGRA	AI.DAS....PDQRT	FYTQLY
AFK83606.1	AANLAAEA....	GTFDA	VHDAARSA	WEERLGLV	RVAQGD....PDDLRT	FYSSTLY
AAO78636.1	SWECA.RWE.K...	MDFED	VKSRTADQ	WEKQLSAI	DVKGGK....KDDRVI	FYTSTLY
AAO79198.1	RENKKAENA.E...	WDFQK	IQNQAEAK	WNHYLGM	IEVEGTN....PDRITQ	FYTHLY
AAO79178.1	ANNFKQELIA.S...	KNFAQ	VKQESD	LWNKELSR	RIISGGT....DDEKTV	FYTSTLY
AAO79070.1	RNNMK.ELD.G...	LTFNEL	LRAKGEAL	WEKELGKY	TLTADR....K.TKET	FYTSAIY
AAO78236.1	RQNMEEQAP.H...	WDFDR	YVAAADAD	WEKQLGKI	EVKGTG....V.QKEI	FYTALY
AAO76985.1	KKNLQAELIP.D...	WNFDG	VKKLAHDE	WNSYLSRM	DVEGTD....D.EKTN	FYTCTY
AAO78878.1	AMNLQAELIS.H...	WDFDK	VVKMSDR	WNKQLDKM	TVESDD....EAAKRV	FYTHLY

	340	350	360	370
AFK83609.1	RMFAMPMNATSTS	GYRGVDG.....	AVH.AAQGFTY	YD
AAO78889.1	RSMLFPRSFYE	IDAKQVVMHY.....	SPYNGKVLPGYMF	TD
AAO77736.1	RSMLFPRSFYE	IDAKQVVMHY.....	SPYNGEVRPGYMF	TD
AAO79067.1	RSVLFPRMFHE	VNAKGETVHY.....	SPYNGEIRPGYMF	TD
AAO79095.1	RSVLFPRKFYE	LDANGQPIHY.....	SPYNGQVLPGYMF	TD
BAA76709.1	RLNLYPNSQF	ENTGTEADPVYRYAS	SPVSATTGPRC.....	
AFK83608.1	RLNLYPNSQF	ENTGTAQEPVYRYAS	SPVSATTGPRC.....	
CAD7619346.1	RLFLYPNSAF	ENVGTTIDPVYKYAS	SPSAATGQDTATTTGAKIVD	GKIYVNNNGF
AFK83605.1	KTLLSPITIGS	DVDGRYRGMDL.....	EVHQAD.GWDY	YQNFSL
AFK83606.1	RSFLAPNVGS	DVDGRYRGWDQ.....	EVHAAEPDFTY	YQNFSL
AAO78636.1	RTYLSPADVSS	PDGAYLGTDG.....	KVY.ISEDFRYYS	NWSL
AAO79198.1	RSFIHPNVCS	DVNGEYMGADF.....	RVHK..SRSKHY	TFSN
AAO79178.1	HTMIDPRIYTD	DVDGRYIGGDK.....	KVHEQDGTFTKRT	IFSG
AAO79070.1	HAALHPFIFQ	SDGQFRGLDK.....	NIE.KAEGFTNY	TVFSL
AAO78236.1	HTMQENTMS	DVNGEYMAADY.....	TTKRVANNETHY	TVFSL
AAO76985.1	HALIQBNQIS	DVDGMYRNAAD.....	SIV.KAGTGTFT	YSTFSL
AAO78878.1	HTMIAETLYC	DVNGEYRGMDN.....	MIY.TDPKKAN	YTLTSL

	380	390	400	410	420	430
AFK83609.1	.F	SVIAYID	PALYRDMVQS	LVYLF	FADAEATGTG	GGLG
AAO78889.1	LF	PFLNLMYPSMNQ	KMQEGLVNAY	YL.E...	S	GFLF
AAO77736.1	LF	PFLNLMYPSMNQ	KMQEGLVNTY	YK.E...	S	GFLF
AAO79067.1	LF	PFLNLMYPSMGE	KMQEGLLNTY	YL.E...	S	GFLF
AAO79095.1	LF	PFLNLMYPSVNKE	KMQEGLINTY	YL.E...	S	GFLF
BAA76709.1		SNPDVTEDLV	DGFVQQYR	.D...	G	GWV
AFK83608.1	AW	PAYSLLYP	ELAAELVD	DGFVQQYR	.D...	G
CAD7619346.1	AW	PAYSLLTP	TFAGELID	GDFVQQYR	.D...	G
AFK83605.1	QATLHALL	PERAQDIV	RSMYQHRV	.E...	G	GWLF
AFK83606.1	QQQLLYLL	APDESAD	MAISLV	RQGG.Q...	G	GWLF
AAO78636.1	KF	LLVLTEP	PAKMRDMATS	LIHL	YATGK...K...	D
AAO79198.1	QI	QLLSMLD	PEVASDIV	ISHQLFAEQ	.S...	G
AAO79178.1	QF	PLQAMIN	PRLVSDALNS	LI	TMA	DQSR...
AAO79070.1	LH	PWFNLVQ	QEVNADIAN	SMLAHY	DKSV...	E
AAO78236.1	SH	PLYTLLE	PERVTD	VKSMIRQ	Y.E...	Y
AAO76985.1	AH	FYTLMP	PERVDD	VNSLIEQ	GE.V...	Q
AAO78878.1	LN	PLMTIIQ	PEMVDN	VINSML	SIYR.Q...	Q

	440	450	460	470
AFK83609.1	AI AKGF DGFDR	LEAYPALQRLV	GQYSAD.....	ELRRGYVAGNFG.....
AAO78889.1	AYIKGLRGYD	IETLWEALKHDA	NAHLR.GTASGRLAYDAYNKL	GYVPNNIGI.....
AAO77736.1	AYIKGLRGYD	IETLWEALKHGA	NAHLR.GTASGRLGYESYNQL	GYVANNIGI.....
AAO79067.1	AFMKNVTKAD	AEKMYEGLLKGAN	NSVHFR.VSTTGRRGYEYNNKL	GYVPYDVKI.....
AAO79095.1	AYMKGVKVDD	IKTLYEGLIHGTEN	VHPE.VSSTGRLGYEYNNKL	GYVPYDVKI.....
BAA76709.1	AYLAGALSTDT	ALEAYDAVVKNA	TVLPA.SNAVGRKGLGQSIFL	GYTEET..T.....
AFK83608.1	AYLKGLPTGT	ALEAYDAALRNA	TVAPP.SNAVGRKGLQTSFFL	GFTPES..T.....
CAD7619346.1	AYLKGVTNFD	VQSFYQSAIRNA	EAVSP.NAGTGRKGLTTSIFD	GYNTS..T.....
AFK83605.1	NFALGTVPD	IADELWDYLVENA	TTTFDDVASVGRRTSTFYAEH	GHVPFPYENEGGLGG
AFK83606.1	AWRQGLLAGH	EEEEAYAVLREN	ADGVFPADSPFNGRAANVEYL	RDGFVHEPARSGKPGD
AAO78636.1	AYRKGTINLD	FRKGYAGMKQEM	ERLPM.....	
AAO79198.1	AYAFGARND	PKPIFKIMRKGA	EPPGSKSQDVETRPGLKQYLDK	GY.....
AAO79178.1	AYMKGIRTYD	VEKAYQAVNTSA	KFGND.....	SLGYTPEP.....
AAO79070.1	MIVKEVKGFD	YERAYEAMKTTA	MNSNYDCLP.....	EYREMGYVPFDKEA.....
AAO78236.1	AILKGIPIGD	MEKAYEAVYNSS	VTISHP.NSPF.....	EVWEKYGFMPENIQT.....
AAO76985.1	AYRKGFRGFD	AERAFNMVKKT	QTVSHPLKSDW.....	EVYTKYGYFPTDLTK.....
AAO78878.1	AYLKGF TGF	D AEEAL TAMKAT	ATYEQNGVP.....	YVMAKGYIPADKIH.....

	480	490	500
AFK83609.1	A SVQRGYDQY	GLSVIAD.ELGL	TEEAETLREQASWPIE
AAO78889.1	GQNVAR TLEYA	YNDWIYTLGK	KLGK.....PAS EIDI FKQRALNYKN
AAO77736.1	GQNVAR TLEYA	YNDWIYTLGK	KLGK.....PEN EIDI YKKHALNYKN
AAO79067.1	NENAR TLEYA	YDWCIIYRMGE	KLGR.....PAE ELLEYKSRSQNYRN
AAO79095.1	NENAR TLEYA	YDWCIIYRLAK	ELKR.....PKKEISLFAKRAMNYKN
BAA76709.1	HE S ASWG	LEGYINDFGIAEMAK	AL SEDPKTPADRVAQLKEATYFEARAEHYVE
AFK83608.1	HE S VSWG	LEGLVNDFGIGNMAA	ALAEDPATPEERRETLRESAAYFLERATHYVE
CAD7619346.1	GEGLAWAMD	GYINDFGIANLAK	ALKEKGDKSDPYANYAADQYQYFLNRAQNYVH
AFK83605.1	QFEERYRHGGS	ATLELALADASL	GAAE.RTGR.....EGGQAFLDKGRNWRN
AFK83606.1	..YDLQHGA	SATMEYALADAML	STMAR.GLGH.....DEDDADRYAARGQSYRN
AAO78636.1	RSPDQKME	SAYDLWAMAKIAE	IGE.....KADSEQYRQRSVSL..
AAO79198.1	YNASI QLEY	TSADFAIGQFALH	AVGD.....EFASWRYFHFARSWKN
AAO79178.1	LSISY TLEYA	YADWCVAQLAK	ALGK.....EEDAKRFYEKGQAYRN
AAO79070.1	ESVSK TLEYA	YDDYCIQAQAK	KL GK.....EDDYHYFLNRALS YQT
AAO78236.1	QSVSI TLEQ	AFDWCVAQLAA	KL NK.....DADYQRFHKKRSEY YRN
AAO76985.1	AESVSS TLES	VYDDYAADMAR	RMGK.....EEDAA YFAKRRADY YKN
AAO78878.1	EATSIAMEY	AVD DWGIAA	MAQ.KMGK.....TADYETFSKRAHY YKN

	510	520	530	540	550	560
AFK83609.1	KLTKPGAWTA	ADGTQVGLLTP	RAADGSW	QSADHAK.....	FEAAGLYQGT	LWQYHWY
AAO78889.1	VYHFKRK...L	MVGKDDK	GVFNFP...KFD	AV.....DWSGE	FCENGS	SWHWSFC
AAO77736.1	VYHFERK...L	MVGKDNK	GVFNFP...NFD	AV.....DWSGE	FCENGS	SWHWSFC
AAO79067.1	LFDPETK...L	MRGKNAD	GTFFQT...PFN	FP.....KWGDA	TEGNS	SWHYTWS
AAO79095.1	LFDKESK...L	MRGRNED	GTFFQS...PFS	PL.....KWGDA	TEGNS	SWHYTWS
BAA76709.1	MFNPEAG...T	FTSRNAD	GTWTSGA.DFD	KK.....AWGGA	TEAS	AWTFGFH
AFK83608.1	LFDPFVD...F	FVPRHED	GTWAVDPET	YDPE.....AWGGG	YETNG	WNFAFH
CAD7619346.1	MFNPSIE...F	FNGRTAN	GAWRSTPD	NFNPA.....VWGS	YETNG	WNMAFH
AFK83605.1	LWNPDVE...L	SGGFQGMVNA	KRPTGEF	VTLPELTD.....	VTRSGF	HEGVPWQYQWM
AFK83606.1	VFDPRTG...N	FRARNAD	GEFFV..GDAD	P.....AHS	DGFHE	GTAVQYQWL
AAO78636.1	LYNS	DTG...W.....LQSR	NPDSGWS	KS L.....GED	FREST	YKNYFWM
AAO79198.1	MFDAEKG...W	FRPRNAD	GSWKAWPEN	A.....LTEE	WYGCIES	NAYQQGF
AAO79178.1	LIDPETK...Y	MRGRDS	KGDWRTPT	FTVA YQGP	GSVHGW	GDITEGFT
AAO79070.1	LFHPKTK...F	FQSKNDK	GEWIEP	FDPIYQYGG...NGG	HPT	TEGNAWQYFWY
AAO78236.1	LFDSQTN...F	MRPRKAD	GTWKS	PFNPSDVGH	A.....EST	GGDYTEGNAWQYTW
AAO76985.1	YFDSSIH...F	IRPKLED	GSWRTPY	DPARS	SIH.....GVGD	FCENGWQYTF
AAO78878.1						

	570	580	590	600
AFK83609.1	DAYDMDALVE	AMGGHEAARL	MRHMG	GEHAPDD.....GKAMLSNAN
AAO78889.1	VFHDPQGLID	LMGGGKEFN	NMDSVF	VIPGKQG..MESRGM
AAO77736.1	VFHDPQGLID	LMGGGKEFN	NMDSVF	VIPGKQG..MESRGM
AAO79067.1	VFHDPQGLAD	LMGGGRKFV	SMMDSV	ENLPIFD.DSY
AAO79095.1	VFHDPQGLID	LMGGKEMF	VIMMDSV	FAPPIFD.DSY
BAA76709.1	APHDQGLA	LYGGGRQGL	LDVMDH	FLTTREKADY...SGI
AFK83608.1	APQDGGGLAN	LYGGQGL	EDKLD	EFSTPEKGA...GNG
CAD7619346.1	VPQDGGGLAN	LYGGKEGL	ATKLD	QFFSTSETGL
AFK83605.1	VPQDVTGLQ	EVMMGGED	GFVERLD	YDYDQPALAA.NP
AFK83606.1	VPQDVPGLF	DLMMGGT	DAAVDRLD	AFAYDELVADPP
AAO78636.1	APQYIDKMI	EWVG.QDS	LRQLTY	FDHHL.....Y
AAO79198.1	VPYDIAGLV	EII	GGKEKA	EKRLDEFF
AAO79178.1	VPHDPVGMV	ELMGGKEE	VIANLT	NLDHTPSDM.....
AAO79070.1	VPQDVQGY	INEA.GKEL	FRKRLD	ELTVELPDDIP
AAO78236.1	VPHNIAQAL	MELTGGT	KAFEQKLD	TFETSTYK.SEQ
AAO76985.1	VQHDVVPGLI	ALF	GGGEP	FLNKLDSL
AAO78878.1	APQDVPYGLI	SIFGG	DKFFT	SKLDSFT

	610	620	630	640	650	660
AFK83609.1	EIDLQAFYLFN	YTFEFSITQ	KWARA	YTKETWNRYI	ATGSSSAVPS	GGGGEFTPPLKTKVY
AAO78889.1	QPIQHMVYLYN	YSSEFWKAQ	HWVREIMDK			LY
AAO77736.1	QPIQHMVYLYN	YSSEFWKAQ	YWIREFIMNK			LY
AAO79067.1	QPIQHMVYLYN	YAGFPWKAQ	YWLREVMNR			LY
AAO79095.1	QPIQHMVYLYD	YAGFPWKAQ	YWLQVMDR			MY
BAA76709.1	QIAHHIFVYLA	EAGDFSGAQ	ELIRIQDR			LF
AFK83608.1	QVSHHIFWLYD	AAGAPSKAQ	EKVREVTTR			LF
CAD7619346.1	QPSHHIAYMYD	YAGQFWKTQ	EKVREALNR			LY
AFK83605.1	EPTIMNAWLYG	YVGPQWKTND	VLAANLNR			F
AFK83606.1	EPNLHAFYVYL	WTQGPWKT	TDVRAASTL			F
AAO78636.1	EPTDHFVYLFN	NRLGAFKTKQ	QIVRSLMTEPMIHKY		GGNSEFKTPYLGKAF	
AAO79198.1	EPSFHFVWIYN	WIGRPYKTE	QEIINRVLNE			QY
AAO79178.1	EPVHFVFFLFN	NQLDVPWYTQ	KWTRYICKN			
AAO79070.1	EPCHHVAYLYN	YLKEFWKCK	KWIRITIVDR			FY
AAO78236.1	EPSHHVAYLYN	FAGQFWKTQ	KYVSHILNT			LY
AAO76985.1	EPSHHVTYLYA	LAGRPERTEL	QELIREIFDT			QY
AAO78878.1	EPSHHIAYLYT	YAGEQWKTA	EKVRFIMSD			FY

	670	680	690	700	710	
AFK83609.1	RLD..FRGMLPTMDN	DAGTMS	TMFVAA	AVGLFPV	TAGSSQFQVGS	PFFDSTTITY....D
AAO78889.1	TAG..FDGYCG..	DEDDNG	QTSAWYV	FSSALGFY	PVCPGT	DQYILGTPLEFKSAKLHL...E
AAO77736.1	TAG..FDGYCG..	DEDDNG	QTSAWYV	FSSALGFY	PVCPGT	DEYIIGTLEFKSAKLHL...E
AAO79067.1	FAT..FDGYCG..	DEDDNG	QTSAWYV	FSSALGFY	PVCPGS	NEVVMGAPYFKKATITL...E
AAO79095.1	TPG..FDGYCG..	DEDDNG	QTSAWYV	FSSALGFY	PVCPGT	DEYVMGTPLFKKATLHF...E
BAA76709.1	VGSDIQGGYFP	..DEDDNG	EFSAWYV	FSSALGFY	PVCPGS	GDYITIGSPLEFDSATLSI...G
AFK83608.1	VGSEIQGGYFP	..DEDDNG	EMSSWVIF	ASLGFY	PLQVGS	DQYAVGSPLEFDDKATVHL...P
CAD7619346.1	IGSAIQGGYS	G..DEDDNG	EMSAWYIL	SAMGFY	PLKMGTFE	YAIIGAPLEFKKATIH...E
AFK83605.1	PDT..FGGGVG	..NDDDLG	TLAAYV	MASLGF	EPVMPGS	GILALNAPKVQAATLT...DAGA
AFK83606.1	TDG..FDGVTG	..NDDDLG	TMSAWH	VSSIGV	YPIVPG	ADLWGLTTPLEFDDVTITL...DPEVF
AAO78636.1	KNA..FEGYS	PEMDEDD	GMSAWYV	FSGAMGFY	PLLVGDEY	YDLTSPLEFDRVLLRL...T
AAO79198.1	SSK..IDGLFG	..NDDDLG	TMGAWYV	FACIGL	YPEIPGV	GFTVNTPIFSSVKVHL...K
AAO79178.1	ANK..VEGIVG	..NDDVG	QMSAWYIL	AASGI	HPSCPGN	TRMEITSPVFDKVEFNLD...SKYH
AAO79070.1	GNT..PDALSG	..NDDCG	QMSAWYMF	NCGIF	YPVAPSS	NIYNIGSPCAEAITVRM...S
AAO78236.1	NNT..SSGYAG	..NDDCG	QMSAWYV	FSSAMGFY	PVNPADGR	YIIGSPLEFDDKATL...A
AAO76985.1	KNK..FDGLCG	..NDDCG	QMSAWYV	FSSAMGFY	PVNPADGR	YIIGSPLEFDDKATL...A
AAO78878.1	TDQ..FDGIIG	..NDDCG	QMSAWYIL	SAMGFY	QVNPSP	DGVFAFGSPREFKKIEVKV...R

	720	730	740	750	760	770
AFK83609.1	DGS	AFVTADGV	SEDAFYVQS	ATLDG	ATFGNTWVD	YATVVG
AAO78889.1	NGK	VTVIKASNN	ITDNR	YVKDM	KVNGK	AFTRNYL
AAO77736.1	NGK	ITVIKADNN	QLDNR	YIKEM	KVNGKS	QTRNFL
AAO79067.1	NGK	KLLEISAP	KNSDAN	YIRSL	NYNGKN	YTKNYL
AAO79095.1	NGK	SLVIDAP	NNSTEN	FYIDSM	SFNGAD	HTKNYLR
BAA76709.1	..GKEL	VINAP	FGASAGE	KYVAGV	SINGAP	ITETTF
AFK83608.1	DGD	..LVVNA	AEENN	SVDNV	VVQS	LAVDGEART
CAD7619346.1	NGK	SIVINAP	NNSKEN	KYVQSM	KVNGK	KAYAKT
AFK83605.1	..TLR	IDAGA				
AFK83606.1	GRD	SLRLT	ADGVAP	DTHTY	TSQSV	LGGEPLDRAW
AAO78636.1	NGN	VLTIQ	TEGR	KKKDA	FIKS	IHFNGKKIADYR
AAO79198.1	KG...	DVIKGG	SEKDI	YIKSM	KNGK	SYESTW
AAO79178.1	QGV	FTIIA	HNNTNN	LYIQK	ALLNGK	EYKNCYLD
AAO79070.1	NGK	NIEMT	ADNWS	PKNL	YVKE	LYVNGK
AAO78236.1	GKKE	FRIR	TIKRS	PEDII	YIQSV	TLNGKKHKDFF
AAO76985.1	DGK	IFTVIA	ENLSKE	HKYVDS	ITLNG	EFPYTKNT
AAO78878.1	GK	VFTV	EAPNN	SKDNI	YIQKV	YLNKFPYHKSY

	780	790	800	810	820	830
AFK83609.1	TAPAF	SMSTAT	DEPAEG	PRVSAE	PTTVQT	GDGGAL
AAO78889.1	EKD	IPYSLS	SFE			
AAO77736.1	EKD	VPYSLS	SFE			
AAO79067.1	ESD	FPYS	SFRDNK			
AAO79095.1	EED	MPYS	SFSKE			
BAA76709.1	DLG	EPLDV		PTTL	VDA	TKVG
AFK83608.1	EDD	APPS	LTG	DE...P	...P	TEVQDATTAG
CAD7619346.1	DQD	ILQS	ITPG	ST...DG	TSLS	PLPLRDVTDR
AFK83605.1	...	...	...	...	...	...
AFK83606.1	PAAS	FGAV	VPADG	...TVE	RL	
AAO78636.1	...	...	...	...	...	...
AAO79198.1	..V	VP	PSFP			
AAO79178.1	SNI					
AAO79070.1	DEA	VP	PSIS	LPEK	...TM	KYKS
AAO78236.1	...	...	...	...	...	...
AAO76985.1	...	...	...	...	...	...
AAO78878.1	SAN	RPIV	LNK			

	840	850	860	870	880	890
AFK83609.1	AASV..VGLPDGV.TAAVTVA..SPTALT	VS	LTGTASADARFFVHLRDAALADGVAAASL			
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	T...PVGSLVDDSMNSKVTFS	GD	TA	AEVVWTSQSGPVS.IGQYTLTSA..AKANAPSSWK		
AFK83608.1	DATTSAAALTDNTSGTRTTT	FAT	TTT	PSITWAGNGIRPT.VG	SYTLTSG.ASGTASPSAWTL	
CAD7619346.1	..EGNGQLLFDNTSNTQLSMKSK	TP	SIVYQFKEGKQN.VKMYTLTSSKASQ	NEDPKSWVL		
AFK83605.1	N.....					
AFK83606.1	FVGATPRQPVLAPGG	RT	E.VAVQVVAQG.....AGTSSGTL			
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1	SIG.....					
AAO78236.1						
AAO76985.1						
AAO78878.1						

	900	910	920	930	940
AFK83609.1	QGQGVSVRSPRLRLSVASA	ERDALAALVDDAVLVRHGNYS	SVTFDRFSTALTKAQEA		
AAO78889.1					
AAO77736.1					
AAO79067.1					
AAO79095.1					
BAA76709.1	EGSLDGTTWTEL.....DS	RS	AQSFSDWTQTRPFTAANQDGYTSVKLT.....LK...TD		
AFK83608.1	EGSDDGETWTTL.....D	ER	SQEQRWALQTRPFTVAEPTAFARYRVT.....VT...AT		
CAD7619346.1	KGSNDGKSWSVL.....D	Q	RKNETFQWRQYTRAFTIQHPGKYSQYKLE.....IT...EN		
AFK83605.1					
AFK83606.1	EVTS	DGAVTATT.....DLAEWT.....			
AAO78636.1					
AAO79198.1					
AAO79178.1					
AAO79070.1					
AAO78236.1					
AAO76985.1					
AAO78878.1					

	950	960	970	980	990	1000
AFK83609.1	LGDEAATSIALRFAADRLGA	AADALDLTGGGYRTLEAEQ	SEAWSGGELKNEANSSSGNLG			
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	A..ATLALSEIELFASA.....S.....A	ADGLS.....				
AFK83608.1	SGSGALSLAEVELLADP.....K.....E	S	GAEEL.....			
CAD7619346.1	AGAEVTTLAELELLGYD.....D.....V	T	NSYQA.....			
AFK83605.1						
AFK83606.1	A.....E	S.....				
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1010	1020	1030	1040	1050	1060
AFK83609.1	GVRSGSWVQYRDMTFETAAGD	TPPRFLT	TVRYDTSFAP	TDTPSTVVRVHAGDVSGP	VVAIVD	
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	IAA.....					
AFK83608.1	TLA.....					
CAD7619346.1	VYEL.....					
AFK83605.1						
AFK83606.1						
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1070	1080	1090	1100	1110	1120
AFK83609.1	LKGTSGWGKYTEVTAELG	VDVQALVDAQVVT	FELLAPSGRSWVG	NFD.WFRFSAEDPA	APG	
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	TETD...ASGYTVTV	DYGDGVPVTDV...		KLTRDDLGGWKVS	APHTFTAPG	
AFK83608.1	VEGD...VAALDVQ	VAFGDGS.EPVA...		GTLRAGAFGGYAV	DAHAHTWTAPG	
CAD7619346.1	NS.L...TTS	LD...HFKKDHKDQAI...		KH...LED	FL...KHLNNKG	
AFK83605.1		EKLPSYVA...		GL		
AFK83606.1		DGLPATVE...		GTVT...	IEAPADAEFG	
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1130	1140	1150	1160	1170	1180
AFK83609.1	QPGESPTVTIEAEDWT...	ASSGRGLKKESST	TWTS	GPVTNVGGTADGD	WIA	YGEVDL
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	TYSA...AVTVRDS	AG...AVAQTTAS	VTAFRDETLVGA...		FNNVCIGDL	
AFK83608.1	VYPV...TVT	VS	GEIETVSASSYVS	SLLREGSLLAA...		YDNVCIGDA
CAD7619346.1	LQD...RISSKAK	GVLSADANQ.LIV	LLARD.....			
AFK83605.1			EVDGVAHTA	AAWLDVA...		ALQD
AFK83606.1	LHTV...RLVVR	DAAGTE...AVREV	SVVVSGESWIADA...		FDNVGIGDA	
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1190	1200	1210	1220	1230
AFK83609.1	PLGELSV....HYVHNS	NRSGNNSALS	SVYLDAFDPANP	GEPF.VTVPLPTTG	SSWTADGT
AAO78889.1					
AAO77736.1					
AAO79067.1					
AAO79095.1					
BAA76709.1	GVTAANCDSQGHG	YFRDKI....NADG	FVQGQTLTIPGST	LT	YDLPAIPAGAPDNITGE
AFK83608.1	GTTV	GSCDGGQGVFFDRAQL....	AAKGFVQGERAT	VP	PGTDLAFDVP
CAD7619346.1					
AFK83605.1	GT.....		LD	FDLS.....	
AFK83606.1	GAANANLDGSG	AYLLRDLL....ADL	GAVQGLELTVPGT	DLTYTLGAPRAG	APDNVAAS
AAO78636.1					
AAO79198.1					
AAO79178.1					
AAO79070.1					
AAO78236.1					
AAO76985.1					
AAO78878.1					

	1240	1250	1260	1270	1280	1290
AFK83609.1	.ATVVLPETVQGT.HEV	FVRLSTEPYADHPY	VANLDSLTFAPGG	PTSVVVESEAWTS	NSG	
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	GQTVKVSAG.AGAKQ	IAFIGTATE....SNR	DSVATLHFTDGST	QQVTISLGDWVG	ASG	
AFK83608.1	GQTI	ELDVP.ADAEQLS	VIGTGTE....KNQ	QATGILTFDDGST	QPIDLSFGDW	SGAAR
CAD7619346.1						
AFK83605.1		GTS.....	AG.....	LTWGTGAA		
AFK83606.1	GEVLEVPEHLRS	SARHLSVVG	TSTH....GTHGG	GLVLGFADGSSQ	TV	DVRLSDWCTGSP
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1300	1310	1320	1330	1340	1350
AFK83609.1	RGLKNESSTWTS	SGPVTINVGGT	ADGDWLAYGEI	DLGSAALDQLS	VHYVHNSNR	SGRNSALS
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	NPYKGN	TV.....	LTISEGRL			SGTGAESS
AFK83608.1	NPVFGNIP		VAVTDSRL			RGGSPQTG
CAD7619346.1						
AFK83605.1	D..R					
AFK83606.1	E..PGNIT		VAKA..GA			RGDRENVQ
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1360	1370	1380	1390	1400	1410
AFK83609.1	VYLDADFDPANP	GPEPFVTVPL	ANTGSSWTTD	GTAVVDLPST	VRGKHQVWV	RSLTEAYADHP
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	VK.....	NTAIYAT		APIDLAVDGD	GAPKVVESLT	MP...KE.
AFK83608.1	.T.....	PAAFFAT		APITLPEGK		RPVSLTLP...DQ.
CAD7619346.1						
AFK83605.1					IP.....	
AFK83606.1	KI.....	GCGLYAT		APVAIPEGK		VLTSVTLP.....
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1420	1430	1440	1450	1460	1470
AFK83609.1	YVANLDSMRFFT	DAYDVEVPPT	DTAALA	AVVDAAGTPE	AEIARYGRID	ARVFTRE....L
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1					TGSL	NDGRVHIFAIASDGDRAA
AFK83608.1					PGELSRDGR	IHVVAVAHDGTFAE
CAD7619346.1						
AFK83605.1						SVS...
AFK83606.1					SDERFHVFAI	ATDAT...
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1480	1490	1500	1510	1520
AFK83609.1	AAARSVLADAG	ATQQAQADER	ARRRLGLATD	QLVPAERRRLE	NL...VASAEAL
AAO78889.1					TDGYSF
AAO77736.1					
AAO79067.1					
AAO79095.1					
BAA76709.1	AAALSVAPGS	VGEQVAGEE	FDAALATVAG	GQG...ETSA	VVNWGDGSPV
AFK83608.1	HPALEVTAAE	GVTLAVGQT	SDVALAQVAG	GREGADLRAA	AVTWGDGSDV
CAD7619346.1					AAGAVTDG
AFK83605.1					SVSG
AFK83606.1					
AAO78636.1					
AAO79198.1					
AAO79178.1					
AAO79070.1					
AAO78236.1					
AAO76985.1					
AAO78878.1					

	1530	1540	1550	1560	1570	1580
AFK83609.1	ESWQAFRTALAAATGTLDDAAASDEALHDARLALQGAVDALEEPADVVLVEVEVSPRCLA					
AAO78889.1						
AAO77736.1						
AAO79067.1						
AAO79095.1						
BAA76709.1	T..HTY.....ATAGAYTVTVTADDGVRSAEAQLQ...MVVVEPEPVYTPETIQVTASAGP					
AFK83608.1	S..HAY.....TAAGTYTAYVVVDDGWTSQVVEVP...VTVTEAEPALAVDVTVSTRCLA					
CAD7619346.1						
AFK83605.1						
AFK83606.1						
AAO78636.1						
AAO79198.1						
AAO79178.1						
AAO79070.1						
AAO78236.1						
AAO76985.1						
AAO78878.1						

	1590	1600	1610	1620	1630
AFK83609.1	GKPYVAV..RAVNVSDAAVDVELASSLGTRSFVG.....VAP.....GASAYQSFAAR				
AAO78889.1					
AAO77736.1					
AAO79067.1					
AAO79095.1					
BAA76709.1	GDTVAVTGAGFAPGEKVSIRIDAETPVSTTAGQNGAIAAELVIPVDAVDGSHPVIAL.GE				
AFK83608.1	GKAYVAV..RAENGEDVFLAIRLVTPFGTKEVAA.....VAP.....GANAYQSFAAR				
CAD7619346.1					
AFK83605.1	GRAFVAV..RATSTADAPVDVTLTTPFGERTVRH.....VQP.....GRSAYQSFTIR				
AFK83606.1	GKAFVAV..RALNTGEQPAAIELATPYGSKLFGD.....VAP.....GANAYQSFAAR				
AAO78636.1					
AAO79198.1					
AAO79178.1					
AAO79070.1					
AAO78236.1					
AAO76985.1					
AAO78878.1					

	1640	1650	1660
AFK83609.1	SATGD.LDVTVTATGADGTQTVEQVVT...VPSCS.....		
AAO78889.1			
AAO77736.1			
AAO79067.1			
AAO79095.1			
BAA76709.1	VSKVEARASVLVATDPQSPRKTSVILAAGSDDAVAGESVPLIATVKPAGAAGKVDFFEGD		
AFK83608.1	VTAVEAGTVTVEATRGTGDEEVTASIQADYAAVTCG.....		
CAD7619346.1			
AFK83605.1	TTSVEAGTATVTIVVAADGT...TSTVDAAYEALACG.....		
AFK83606.1	AAAVEAGEVTVTVITPDGE...PQQVTAAYDAAACS.....		
AAO78636.1			
AAO79198.1			
AAO79178.1			
AAO79070.1			
AAO78236.1			
AAO76985.1			
AAO78878.1			

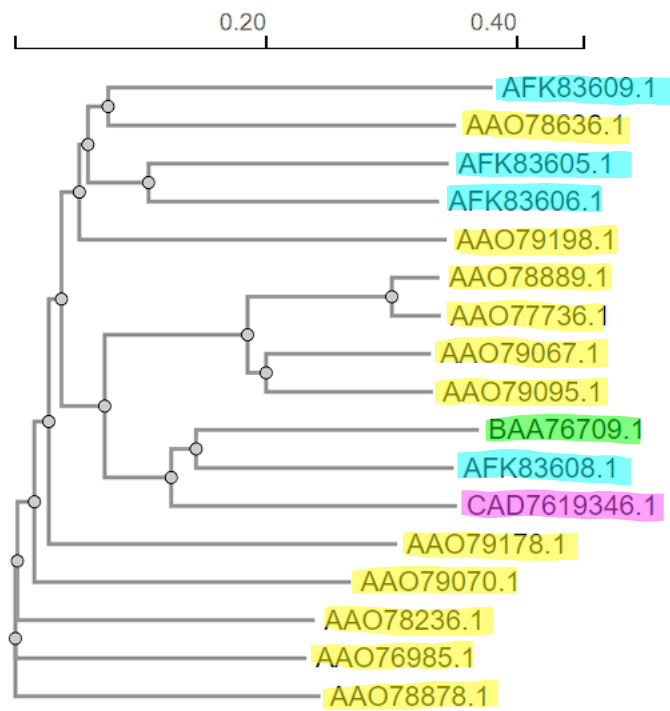
AFK83609.1	
AAO78889.1	
AAO77736.1	
AAO79067.1	
AAO79095.1	
BAA76709.1	EIVGSAEVTGGTATADVTTIPTSGTHVYVASFVPSDETAYRGSESSPLEIEVRDTPVLQAE
AFK83608.1	
CAD7619346.1	
AFK83605.1	
AFK83606.1	
AAO78636.1	
AAO79198.1	
AAO79178.1	
AAO79070.1	
AAO78236.1	
AAO76985.1	
AAO78878.1	

AFK83609.1	
AAO78889.1	
AAO77736.1	
AAO79067.1	
AAO79095.1	
BAA76709.1	LVIGASTVTQGGTVQLAGRGFGSGETISVTLHSDPIHLIDVVADGNGAFRVTVTIPANAE
AFK83608.1	
CAD7619346.1	
AFK83605.1	
AFK83606.1	
AAO78636.1	
AAO79198.1	
AAO79178.1	
AAO79070.1	
AAO78236.1	
AAO76985.1	
AAO78878.1	

AFK83609.1	
AAO78889.1	
AAO77736.1	
AAO79067.1	
AAO79095.1	
BAA76709.1	VGAHTLIAVGADSGTLTAQGALTVTAAAGTGGGLAGTGGAVPIAAIALLLLLLLAAGGVLVI
AFK83608.1	
CAD7619346.1	
AFK83605.1	
AFK83606.1	
AAO78636.1	
AAO79198.1	
AAO79178.1	
AAO79070.1	
AAO78236.1	
AAO76985.1	
AAO78878.1	

AFK83609.1	
AAO78889.1	
AAO77736.1	
AAO79067.1	
AAO79095.1	
BAA76709.1	RRRRVES
AFK83608.1	
CAD7619346.1	
AFK83605.1	
AFK83606.1	
AAO78636.1	
AAO79198.1	
AAO79178.1	
AAO79070.1	
AAO78236.1	
AAO76985.1	
AAO78878.1	

Fylogenetický strom:



Bacteroides thetaiotaomicron

Cellulosimicrobium cellulans

Microbacterium sp.

Neobacillus novalis